

Executive MBA Financial Accounting

TA Review Session 4

Shirley Xu

CHICAGO BOOTH



The University of Chicago Booth School of Business

Reminders...

Contacts

- **Shirley Xu**, room G11, xxu32@alumni.chicagobooth.edu
- **Angela Cheng**, room G12, hcheng12@alumni.chicagobooth.edu
- Assignment 5 is due tomorrow
- Assignment 6 during non-study weeks

Office Hours

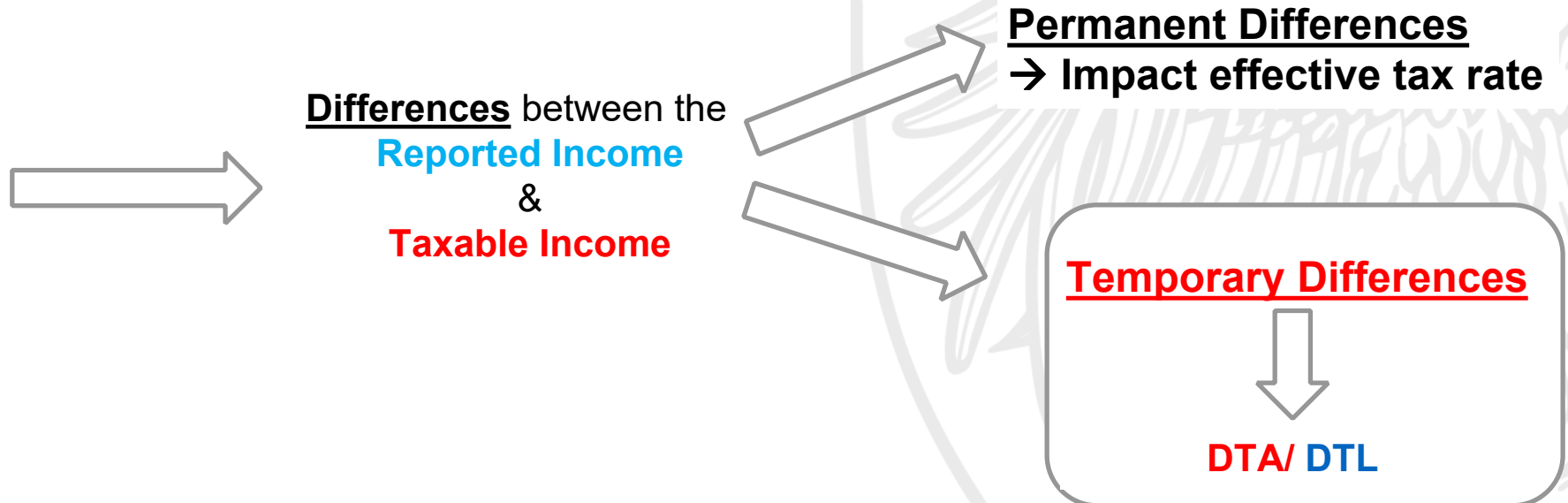
- Optional, during non-class weeks, sessions will be held on Mondays
 - 10/3, 17/3, 24/3, 31/3 – 8:00-9:00 PM HK time
- Please submit questions in advance via emails (until the day before)

Deferred Tax Assets & Liabilities (DTA & DTL)



Why? What? How?

- Tax Book is different from Reported Book
- Tax rules vs. GAAP rules
- Book Revenue vs. Taxable Revenue
- Book Expense vs. Tax Deductible Expenses



Examples of the two types of tax differences?

Temporary Differences

- e.g., depreciation, bad debt expenses
- A revenue or expense item that affects book income and taxable income at different times, where the temporary difference will eventually *reverse*

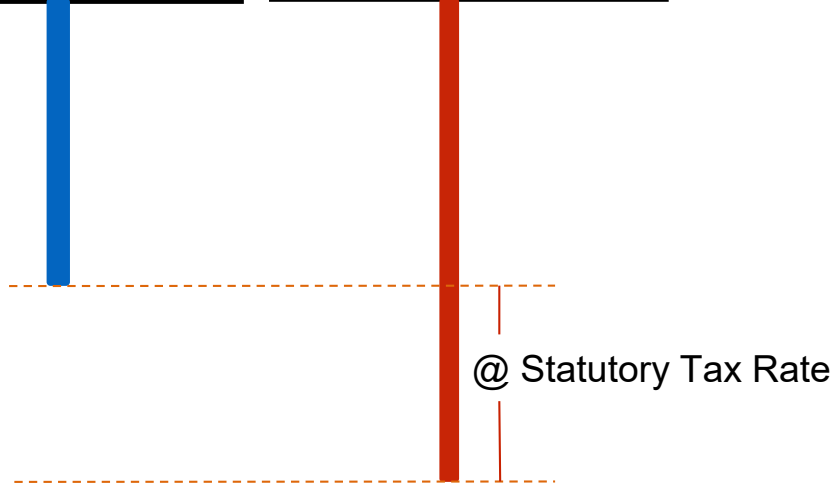
Permanent Differences

- e.g., fines, non-taxable income
- A difference between book income and tax income that will *not reverse* in a subsequent period

DTA or DTL?

Arising only from Temporary Differences

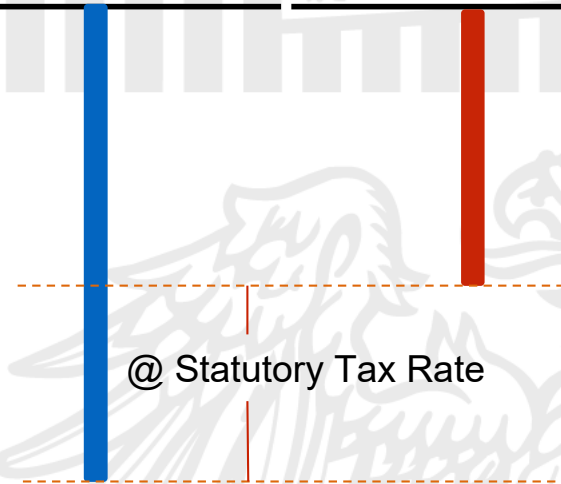
Book Income Taxable Income



Less book rev. / More book exp.
 Book Income < Taxable Income
 Tax paid now > Tax paid in the future

➡ **DTA (tax benefit)**

Book Income Taxable Income



More book rev. / less book exp.
 Book Income > Taxable Income
 Tax paid now < Tax paid in the future

➡ **DTL (tax liability)**

* Over the years, the total amount of tax paid is the same

Journal Entry

Interest Expense = Tax Owed/ Paid + **DTL**

OR

Interest Expense = Tax Owed/ Paid - **DTA**

Dr. Interest Expense

Dr. DTA

Cr. DTL

Cr. Tax owed/ Paid
(Tax Payable/ Cash)

Calculations

- **Taxable Income * Statutory Tax Rate (STR)
= Taxes Owed/ Paid**
- **Temp Book–Tax Income Difference x STR = DTA/ DTL**
- **DTA or DTL / STR = Book–Tax Differences**
- **Income Tax Expense / Book Income Before Tax
= Effective Tax rate (ETR)**
- **If only Temp. Book-Tax Income Diff → ETR = STR**

Permanent Differences

- **No DTA/DTL**

- Because a permanent difference will never reverse

Impact on Effective Tax Rate (ETR)

- $ETR = \text{Income Tax Expense} / \text{Book Income (before Taxes)}$
- $ETR \neq STR$ because of permanent differences

➡ Permanent Tax difference

$= (ETR - STR) * \text{Book Income Before Tax}$

Review Question: Deferred Tax Assets & Liabilities



Review Question: Deferred Taxes

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
- Income tax expense = \$156,000
- Taxes owed = \$48,000
- Statutory tax rate = 40%

Q1: Calculate Taxable Income for FY 2024

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
 - Income tax expense = \$156,000
 - Taxes owed = \$48,000
 - Statutory tax rate = 40%
- Formula: $\text{Taxable Income} * \text{STR} = \text{Taxes Owed}$
 - Plug in numbers: $\text{Taxable Income} * 40\% = 48,000$
 - $\Rightarrow \text{Taxable Income} = 48,000 / 40\% = \$120,000$

Q2: Show journal entries for Income Tax Exp. \$156,000; Assume all taxes owed are accrued.

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
- Income tax expense = \$156,000
- Taxes owed = \$48,000
- Statutory tax rate = 40%

Dr. Income Tax Expense	156,000	
Cr. Taxes Payable		48,000
Cr. Net Deferred Tax Liability		108,000

Q3: Calculate Effective Tax Rate for FY 2024

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
 - Income tax expense = \$156,000
 - Taxes owed = \$48,000
 - Statutory tax rate = 40%
- Formula: $ETR = \text{Income Tax Expense} / \text{Income before Taxes}$
 - Plug in numbers: $ETR = 156,000 / 318,000 = 49.06\%$

Q4a: \$ Amount of temporary difference between book income and taxable income?

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
- Income tax expense = \$156,000
- Taxes owed = \$48,000
- Statutory tax rate = 40%

Remember that only temp diff can give rise to DTA/DTL!

- DTL = 108,000 (increase in DTL)
- Income subject to temporary tax difference
- $= 108,000 / 40\% = 270,000$

Q4b: \$ Amount of permanent difference between book income and taxable income?

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
- Income tax expense = \$156,000
- Taxes owed = \$48,000
- Statutory tax rate = 40%

Another way: $156,000 - 48,000 * 40\% = 28,800$

- Permanent tax difference = $(ETR - STR) * \text{Book Income}$
- $= (49.06\% - 40\%) * 318,000 = 28,800$
- Income subject to permanent tax difference
- $= 28,800 / 40\% = \$72,000$

Q5: Use Q4, Reconcile difference btw. Book Income (318,000) and Taxable Income from Q1

The following information is extracted from the annual report of Haresh Limited (the company) for fiscal year 2024:

- Income before taxes (Book income) = \$318,000
- Income tax expense = \$156,000
- Taxes owed = \$48,000
- Statutory tax rate = 40%

First step: figure out the numbers at the “income” level

- Income subject to temporary tax differences
- $= 270,000 = (156,000 - 48,000) / 40\%$
- Income subject to permanent tax differences
- $= 72,000 = (49.06\% - 40\%) * 318,000 / 40\%$

Q5: Use Q4, Reconcile difference btw. Book Income (318,000) and Taxable Income from Q1

From Q1 & Q4, we already know that:

- Book Income before taxes = \$318,000 (given)
- Income subject to temporary difference = \$270,000 (Q4a)
- Income subject to permanent difference = \$72,000 (Q4b)
- Taxable income: \$120,000 (Q1)

Second step: What are we trying to do here?

So, the question is essentially asking us:

- Income before taxes (book income): \$318,000
- Plus or minus? income subject to temp diff: \$270,000
- Plus or minus? income subject to perm diff: \$72,000
- Equals to taxable income: \$120,000

Q5: Use Q4, Reconcile difference btw. Book Income (318,000) and Taxable Income from Q1

Answer: $318,000 - 270,000 + 72,000 = 120,000$

Why do we minus income subject to temp diff 270,000?

- Temp diff gives rise to a DTL of 108,000
- Meaning taxable income < book income in this period
- So from book income → taxable income, we need to subtract

Why do we plus income subject to perm diff 72,000?

- Effective tax rate (49.06%) is higher than statutory one (40%)
- Meaning overall we pay more tax → there are net expenses included in book income that are not tax-deductible
- So from book income → taxable income, we need to add

Q & A

